



Optimization of short micro pin fins in minichannels

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ABSTRACT

Finned minichannels are modeled in order to optimize microstructure geometry and maximize heat transfer dissipation through convection from a heated surface. Six pin fin shapes – circle, square, triangle, ellipse, diamond and hexagon – are used in a staggered array and attached to the bottom heated surface of a rectangular minichannel and analyzed. Also, using square pin fins, different channel clearance over fins are investigated to optimize the fin height of the fins with respect to that of the channel. Fin width and spacing are investigated using a ratio of fin width area to the channel width. Fin material is then varied to investigate the heat dissipation effects. Triangular fins with larger fin height, smaller fin width, and spacing double the fin width maximizes the number of fins in each row and yields better performance. Correlations describing the Nusselt number and the Darcy friction factor are obtained and compared to previous ones from recent studies. These correlations only apply to short fins in the laminar regime. Completely understanding the effects of micro pin fins in a minichannel is essential to maximizing the performance in small scale cooling apparatuses to keep up with future electronic advancements.

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1. Introduction

Cell phones, computers, and MP3 players, are only a few devices which express the importance of electronics in today's society. There is a drive to obtain the most efficient and powerful yet more compact electrical devices, and engineers are trying to increase the threshold that thermal effects have on the component material. Overheating of these electrical components are a concern as the temperatures reach values that threaten the proper functioning and their physical integrity. Currently, electrical components must sustain a low constant surface temperature to avoid overheating. The advancements in the electrical devices are limited to the absence of the efficient methods to remove the heat that is being generated.

Small channels have been effective in removing heat through convection from the surface of a microchip. These channels act as heat exchangers at the nano, micro and miniscales. Forced and natural convection from fluids flowing through these channels can dissipate high surface temperatures. Nano, micro, and minichannels have a higher heat transfer surface area to fluid volume ratio than a conventional channel which enhances convection. The heat transfer coefficient increases as the hydraulic diameter's size is reduced in the channel enabling an excellent cooling apparatus. Although it has excellent cooling capabilities, these channels experience a high pressure drop as fluid flows. This can cause problems when trying to re-circulate the fluid with a pump [1]. Because of

the innovations in electrical devices, optimizing these channels has been an important part of the research industry.

Modifications to mini/microchannels to enhance thermal performance are being examined by many researchers. Data shows that increasing surface roughness [2,3] and/or applying small cavities [4–8] on the channel walls will improve the heat transfer. The addition of fins or small cavities on the surfaces of the channel creates small disturbances within the fluid which allow fluid to mix and enhancing heat transfer.

Fins made up of different shapes, sizes, and materials can be placed in different arrangements to increase the channel surface area and affect the heat transfer performance and pressure drop characteristics [9]. Many modifications to the microchannel's surface have been investigated in enhancing the thermal performance. A small review of some of the work in single phase heat transfer enhancement techniques in mini/microchannels and microdevices are discussed in [10,11]. Zhong et al. [12] examined the effects of varying the thermal conductivity of the fins structures on the surface of a microchannel. With increasing thermal conductivity, temperature decreases with little effect on pressure drop. Lee et al. [13] investigated a microchannel with silicon based oblique fins and compared it with an unfinned channel and obtained slight increase in heat transfer performance with little or no increase in pressure drop. John et al. [14,15] computationally investigated the effects of fin shape and found that the optimal shape of the fin was dependent on the flow rate used.

Peles et al. [16] analytically completed a heat transfer analysis over a bank of cylindrical micro pin fins. It was concluded that at high Reynolds number, Re , a denser pin fin configuration was more

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Nomenclature

A_b	surface area of the heated base region	Nu	Nusselt number
A_c	flow area of channel	Nu_f	Nusselt number across the fins
A_f	cross sectional area of fin	P_c	wetted perimeter of channel
A_{fin}	surface area occupied by the fins	P_f	wetted perimeter of fin
a, b, c, e, r, s	fin geometrical dimension	p	pressure
$C_{Nu}, C_{f,S}, U, V, W, X, Y, Z$	constants	Pr	Prandtl number of fluid
c_p	specific heat of constant pressure	Pr_s	Prandtl number at surface
d_c	depth of channel	q''	applied heat flux
dh	clearance of channel ($h_c - h_f$)	Re	Reynolds number
d_h	depth of heated surface	Re_f	Reynolds number of the fin
D_c	hydraulic diameter of channel ($4A_c/P_c$)	S_L	longitudinal pitch (center to center)
D_f	hydraulic diameter of fin ($4A_f/P_f$)	S_t	transverse pitch (center to center)
f	Darcy friction factor	T	temperature
h	average heat transfer coefficient	T_{nw}	near wall temperature
h_b	average heat transfer coefficient of base	T_w	wall temperature
h_c	height of channel	u	axial velocity
h_{fin}	average heat transfer coefficient of fin	u_{max}	maximum axial velocity
h_f	height of fins	w_c	width of channel
k	thermal conductivity	w_f	width of fins
M	number of velocities used; $M = 5$		
MAE	mean absolute error		
N	total number of fins		
N_x	number of fins along the x axis		
N_y	number of fins along the y axis		

Greek symbols

μ	dynamic viscosity
ρ	density

efficient, and at low, Re , a more sparse arrangement was recommended. Selvarasu et al. [17] and Shafeie et al. [18] also examined the density of cylindrical fins minichannel. When measuring heat capacity and pressure drop, channels with fins of lower density provided the best performance in the laminar flow however the effects of the increase pressure drop greatly outweighed the increase of heat removed. Koz et al. [19,20] also computationally completed a parametric study using cylindrical micro pin fins to evaluate the effects of fin height versus fin diameter and the longitudinal and transverse distances to diameter ratios.

Reyes et al. [21] investigated the effects of fin clearance of microchannels and found that there was a decrease in thermal performance; however, less power to pump the fluid into the system was needed. Min et al. [22] found that with fins about 60% of the channel height yields less resistance and improves cooling performance.

Heat transfer and pressure drop characteristics for liquid single phase flow over an array of micro pin fins in a minichannel are investigated in [23–27]. Correlations of Nusselt number, Nu , with respect to Re and of Darcy friction factor, f , in order to obtain the pressure drop, were obtained and compared to previous work. Kosar et al. [28] obtained a friction factor correlation across a bank of low aspect ratio pin fins. Vanapalli et al. [29] investigated the pillar ‘fin’ shape which contains the lowest friction factor using nitrogen gas flowing through the microchannel. Various geometries were tested in staggered arrangements and found that pillars with the sine shape cross-sections provide the lowest friction factor. Kosar et al. [30] further investigated the effects of pin fin aspect ratio, fin configuration, fin spacing, and fin shape for pressure drop. It was concluded that the existing conventional correlations cannot accurately capture the microscale interaction of the fluid and the fins.

All of these studies have demonstrated some progress in the development of enhancing cooling for electrical devices but the optimum fin material, shape, and geometry still need to be determined.

With the increasing desire to improve the cooling techniques for small electrical components, advanced materials with high thermal properties need to be exploited. Fins made up of silicon or carbon nanotubes (CNTs), both containing high thermal

properties, have been the choice of materials to improve thermal performance. CNTs have a unique molecular structure that contains excellent physical and mechanical properties including great strength, chemical stability, and high thermal conductivity. Thermal conductivities of CNTs have been calculated to be as high as 6000 W/mK-higher than diamond at 2000–2500 W/mK and graphite at 2000 W/mK [31,32]. There are ongoing investigations in finding an effective thermal conductivity when clumping CNTs together to form a fin. Assembling the fins together decreases the thermal conductivity and it is unknown what the relation is to calculate that value. Some say it is as high as 400 W/mK [33–35]. CNTs can be grown directly on the channel surface in mass quantities using chemical vapor deposition. There are a few of ways to get the desired shapes of fins. One is a metal catalyst is placed on the surface in the desired shapes of the fins and grown directly on the surface. Another way to form the shaped fins is to coat the surface for a given area with CNTs and laser cut fins to obtain the desired geometry. Both processes for growing fins on the surface of a micro or minichannel are further described in [35–37].

Microstructures can be placed in a staggered or inline formation along the surfaces of the walls. These fins can be different shapes and sizes and can be placed in various patterns in order to improve the thermal performance. Shape of the fins affect the motion of flow by producing a mixing in the flow as the fluid passes through the channel, improving the performance. Optimizing the fins shapes, topology, and dimensions to achieve the maximum thermal performance has not been thoroughly executed with corresponding heat transfer and friction factor correlations. In expanding on the work conducted by Tullius et al. [38], this paper discusses the effects of CNT micro fins on the surface of a minichannel in a staggered formation. In order to completely optimize the fins used, a few comparisons are conducted. Different pin fin geometries are modeled and compared for various flow rates ranging from 0.0625–1 m/s and inputted heat fluxes ranging from 10 to 150 W/cm² in the single phase regime. Optimum fin height, width, and spacing of the square shaped pin fin are then modeled. Also, fin material properties are varied to observe the affects. All these simulations are compared to the base unfinned channel to observe the

enhancement obtained with the microstructures. Results are provided in terms of average Nu versus Re and pressure drop versus axial velocity. Nu and friction factor correlations then are obtained and compared to previous studies.

2. Computational modeling

For these simulations the flow regime is considered to be a continuum flow. The flow through this model can be solved using macroscopic relations or the Navier Stokes equations. The basic governing equations for a steady-state, incompressible flow are: Continuity equation

$$(\nabla \cdot \rho u) = 0 \quad (1)$$

Conservation of momentum

$$\rho(u \cdot \nabla u) = -\nabla p + \mu \nabla^2 u \quad (2)$$

Conservation of energy

$$\rho c_p (u \cdot \nabla T) = k \nabla^2 T \quad (3)$$

The geometry is meshed based on the finite volume method. In this technique, the region is divided into small sub-regions known as control volumes. Approximations of the values solved by the Navier Stokes equations are obtained for each control volume. Combining the control volumes, the values display the behavior of the whole region as an entity profiling the fluid flowing through this rectangular channel. ANSYS CFX, based on the Navier Stokes fluid flow equations, is used to model the flow past the pin fins in a staggered array in the minichannel and solves the governing equations iteratively for each control volume. The accuracy of the solution is proportional to the size and shape of the control volume and the size of the final residuals. The convergence criterion for the solution is residuals of less than 10^{-6} for both continuity and momentum. Each model contained a fine mesh surrounding the finned section with a less dense mesh as the channel extends to the inlet and outlet.

Four different comparisons of finned minichannels to the no finned channel are completed in this paper in order to optimize the micro-pin fins using multiple flow rates and heat fluxes: (1) six different pin finned shapes, (2) varying height spacing of fins square pin fins, (3) fin width and spacing, and (4) applying multiple material properties to the square pin finned geometry. A rectangular minichannel is modeled with dimensions, $15 \times 1 \times 45 \text{ mm}^3$ ($w_c \times h_c \times d_c$). Micron sized structures, acting as pin fins to increase the surface area, are placed along a $15 \times 25 \text{ mm}^2$ ($w_c \times d_h$) area on the bottom surface of the channel in a staggered array with a height, h_f , and a width, w_f . A constant heat flux is applied to the bottom of the channel. This is shown in Fig. 1. The dimensions and shape of the channel were chosen for simplicity of the model as well as ensuring the flow is hydrodynamically developed before the fluid reaches the heated region.

Six geometrical shapes are investigated for the fins: circle, square, triangle, ellipse, diamond and hexagon. They were chosen to get the effects of a variety of shapes that may or may not be the most aerodynamic in nature. The dimensions and shapes modeled in this study are shown in Fig. 1b. The transverse and longitudinal spacing of the fins, S_t and S_L , for this study is equal to double the width of the fin, $2w_f$. Fin height, h_f , is half the length of the channel height, h_c , or 0.5 mm. A staggered formation is proven to provide an enhanced heat transfer compared to an inline array due to more mixing within the fluid. Therefore, to obtain a staggered formation, the rows are offset by w_f . For all geometries, the fin material property used is that of CNTs; however, because the thermal conductivity is unknown when nanotubes are clustered together to form fins, an effective thermal conductivity of 400 W/mK is used [12]. This

study was simulated using multiple heat flux and flow rate inputs. The next comparison uses the square geometry for the fins. Using the square geometry and CNT material properties as the first study, channel clearance is varied using fin to channel height ratios, h_f/h_c between 0.25 and 0.75. Channel clearance can be defined as the subtraction of the fin height from the height of the channel. A side profile of the fins relative to the channel is shown in Fig. 1c. This study is designed to understand the optimum fin height with respect to the channel and is compared to the channel with no fins.

The third investigation was conducted in efforts to optimize the fin dimensions and spacing using the square pin fin geometry. Because the fin width, w_f , fin spacing, S_t , S_L , and number of fins, N , are all dependent on one another, the ratio of fin width times the number of fins in the row, N_x , to the channel width, $N_x w_f/w_c$, is kept constant at a value of 0.5. The spacing S_t and S_L are kept equal to double the fin width. The height of the fins to the channel height is kept at half the channel clearance and the number of fins along each row varies based on the width and the spacing of the fins. Fin width is varied from 0.5 to 2.5 mm resulting in 15–3 fins per row, respectively. Once the optimum size of fins is determined, the spacing is varied using the fin's width of 1.25 mm. Unlike the previous spacing that is the same as the fin width, the spacing is reduced to 2 mm and 2.25 mm while maintaining the same number of fins in the row. These were all compared to the smooth, unfinned channel.

The fourth investigation is performed to understand how fin material properties enhance heat transfer of minichannels. The same square pin fin geometry is used similar to the first study with the fin width at 1 mm, spacing at 2 mm, and the fin height half of the channel height. Because the effective thermal conductivity of CNTs grouped together to form fins is unknown, the thermal conductivity is differed with values ranging from 200 to 800 W/mK. In addition to varying the thermal conductivity of CNTs, silicon, copper, and aluminum are also used as the fin material. Table 1 has a list of properties pertaining to the different fin materials. Table 2 summarizes the modeled simulations for each study giving the shapes and dimensions used for each simulation.

Initial inlet temperature and outlet static pressure values applied to the model are assumed for all simulations to be 25 °C and 0 Pa, respectively. To monitor the heat transfer coefficient and the heat transfer rate, the outer walls of the channel are set to be adiabatic. No-slip boundary conditions and no interfacial resistance are assumed at the wall/fluid interface. Water is used as the working fluid flowing through this heat exchanger with velocities ranging from 0.0625 to 1 m/s through the inlet of the channel. These simulations are in the single phase regime and fluid properties are kept constant throughout the simulations. Water flows past the pin fins carrying heat subjected by the bottom surface. A constant heat flux ranging from 10 to 150 W/m² is applied to a $15 \times 25 \text{ mm}^2$ area at the bottom of the channel. With the constant heat flux, the different geometry of the nanotube bundles, and the forced convection, the temperature across the surface of the microchip and the bulk temperature of the fluid vary. To accurately obtain the heat transfer coefficient, h , across the microchip region for the simulations, an average h was obtained.

There were some uncertainties within the model that can increase the error in the approximation. No slip boundary conditions are used to define these simulations at the fluid solid interfaces. As the scales start moving toward the micro or nanoregime, the conventional continuum calculations can no longer be used. Also, the resistance created at the interface of the CNTs and the working fluid, water, is still a major issue. In this model, no interfacial resistance is applied. The CNT fins are modeled as a solid emerging out from the surface rather than bundles of nanotubes. CNT micro pin fins are made up of many nanotubes where the fluid penetrates through the small gaps, or nanochannels, allowing for an increased

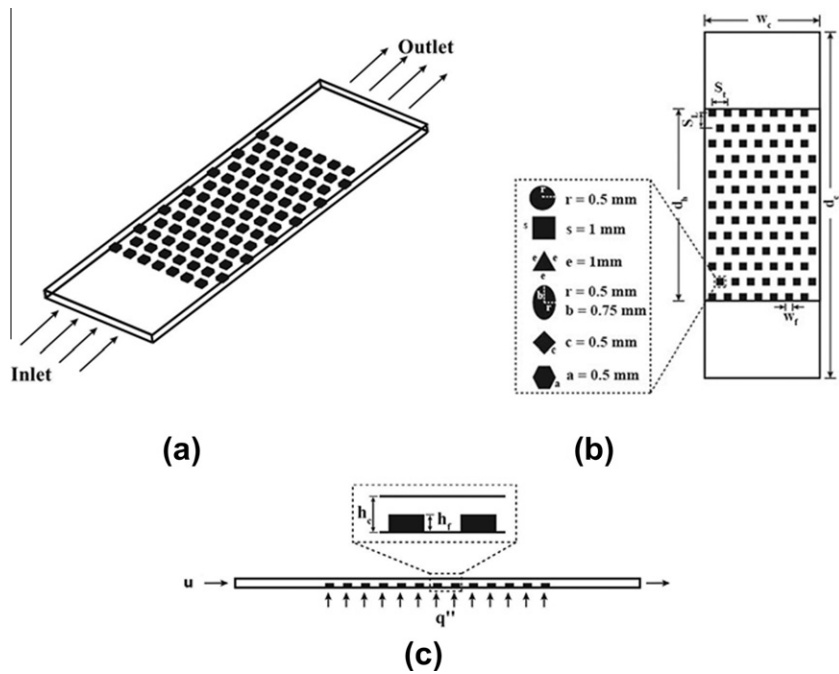


Fig. 1. Minichannel model with dimensions: (a) Isometric view of minichannel; (b) Fins geometry for first investigation is made up of six different shapes and compared to a no finned channel; and (c) Side view of the channel with magnified view of fins relative to the channel height.

Table 1
Material properties for fin material comparison.

Fin material	k (W/mK)	cp (J/kg K)	ρ (kg/m ³)
Aluminum	237	903	2702
CNT	200–800	450	1300
Copper	401	385	8933
Silicon	148	712	2330

in convection within the system. As shown in Shenoy et al. [37], CNTs absorb the fluid at high temperatures creating a porous like material and increases heat transfer due to a hastened nucleate boiling onset initiating phase change. This assumption in the model may underestimate the thermal performance of CNT fins. Also,

there are ongoing investigations of the effective thermal conductivities of CNT bundles; therefore, an estimation of an effective thermal conductivity is used. There are many assumptions made within this program that increases the error of the approximation, however, the general trend should be similar.

3. Results and discussion

This work was conducted to understand how the shape, topology, and the material properties of micro pin fins on the surface of a minichannel can affect thermal performance. Four different studies were conducted to optimize the minichannel in order to obtain the maximum amount of heat transferred. The results are presented by both comparing two non-dimensional units, average

Table 2
Summary of simulations performed.

Study	Shape	Fin material	w_f (mm)	N_x	N	$\frac{N_x \times w_f}{W_c}$	h_f (mm)	$\frac{h_f}{h_c}$	S_t (mm)	S_L (mm)
1	Square	CNT	1	7	91	7/15	0.5	50	2	2
	Triangle	CNT	1	7	91	7/15	0.5	50	2	2
	Ellipse	CNT	1	7	91	7/15	0.5	50	2	2
	Diamond	CNT	1	7	91	7/15	0.5	50	2	2
	Hexagon	CNT	1	7	91	7/15	0.5	50	2	2
2	Square	CNT	1	7	91	7/15	0.25	25	2	2
	Square	CNT	1	7	91	7/15	0.33	33	2	2
	Square	CNT	1	7	91	7/15	0.5	50	2	2
	Square	CNT	1	7	91	7/15	0.66	66	2	2
	Square	CNT	1	7	91	7/15	0.75	75	2	2
3	Square	CNT	0.5	15	375	1/2	0.5	50	1	2
	Square	CNT	0.75	10	170	1/2	0.5	50	1.5	1.5
	Square	CNT	1.25	6	72	1/2	0.5	50	2	2
	Square	CNT	1.25	6	72	1/2	0.5	50	2.25	2.25
	Square	CNT	1.25	6	66	1/2	0.5	50	2.5	2.5
	Square	CNT	1.5	5	40	1/2	0.5	50	3	3
4	Square	CNT	2.5	3	15	1/2	0.5	50	5	5
	Square	Silicon	1	7	91	7/15	0.5	50	2	2
	Square	Copper	1	7	91	7/15	0.5	50	2	2
	Square	Aluminium	1	7	91	7/15	0.5	50	2	2

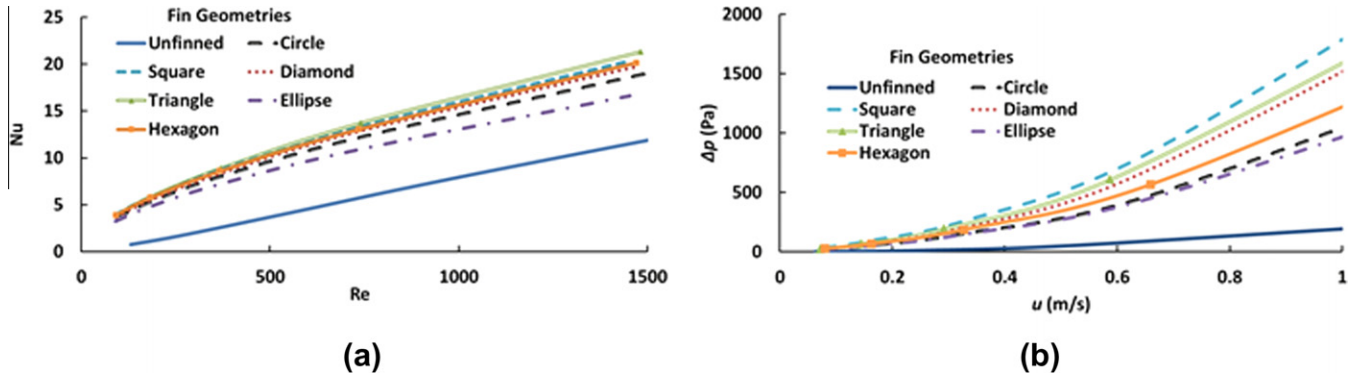


Fig. 2. Results for different fin geometry: (a) Nusselt number, Nu, versus Reynolds number, Re; (b) Pressure drop, Δp , across the micro pin fins versus maximum axial velocity, $u_{\max}$.

Re and Nu, and by relating change in pressure, Δp , and axial velocity, u . Average Re and Nu are collected for the different inputted heat fluxes and flow rates. The Re is defined as the multiplication of the density, ρ , average axial velocity, u , and the hydraulic diameter of the channel, D_c , over the dynamic viscosity, μ . Average axial velocity was obtained as the fluid flowed across the fin bank.

$$Re = \frac{\rho u D_c}{\mu} \quad (4)$$

Nu is proportional to the average heat transfer coefficient, h , and the hydraulic diameter of the channel, and non-proportional to the thermal conductivity, k .

$$Nu = \frac{h D_c}{k} \quad (5)$$

The average heat transfer coefficient is obtained by the amount of surface area of the fins and the base that the fluid interacts with. The expression is given below.

$$h = \frac{(h_{fin} A_{fin} + h_b A_b)}{(A_{fin} + A_b)} \quad (6)$$

h_{fin} and h_b are the heat transfer coefficients of the fin and the base of the heated region only; and A_{fin} and A_b are the surface areas in which the fluid touches the fin and the heated base region, respectively. The heat transfer coefficient of the fin and the base at the fluid interface is obtained by

$$h = \frac{q''}{T_w - T_{nw}} \quad (7)$$

where q'' is the wall heat flux, T_w is the temperature of the wall and T_{nw} is the near wall temperature. Pressure drop is determined

across the pin fins only to avoid any entrance and exit effects of the fluid flow using a maximum axial velocity. The results for all of the studies are given below.

3.1. Different shaped micro fins

For the first investigation, results of different finned shaped microstructures on surfaces of minichannels are compared to the smooth, unfinned channel. With increasing Re, Nu values increase for all minichannel simulations. The Nu for the triangle, diamond, square and hexagon shaped fins have similar values with a maximum average Nu difference of $\approx 12\%$ for higher Re. For this data, triangular shaped fins showed better thermal performance. The circle and ellipse shaped portrayed a thermal performance is less effective than the triangle shaped fins by about 22% and 37%, respectively, however, still more efficient than the unfinned channel. For lower Re, the Nu increase is very high compared to the unfinned channel and at higher Re the gap decreases to have a 60% and 45% performance enhancement for the circular and ellipse shaped fins respectively. Fig. 2a and b reflect the results of this study.

With the addition of fins, there is an increase of pressure drop. A value too high can create problems for the pumps to flow fluid through the minichannel. As the velocity increases, the pressure drop increases for all models. The square pin fin curve has the highest pressure drop with up to about 2.5 kPa followed by the next cluster of shapes: triangle and diamond. This increase is much higher than the unfinned channel which yields a max pressure drop of 0.2 kPa for the same Re. Circular shaped fin gives a max pressure drop of about 1.6 kPa for the highest flow rate. The ellipse shape shows a smallest pressure drop just under the circle shaped

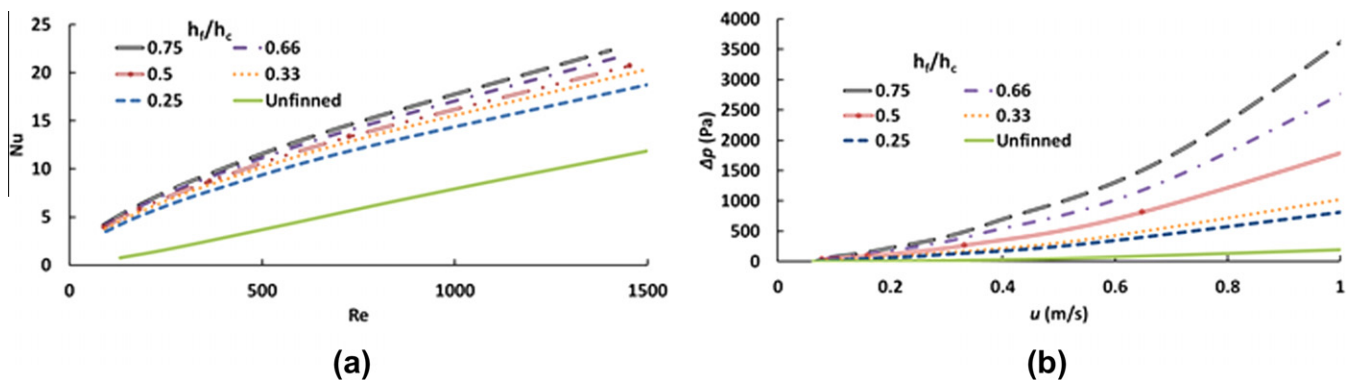


Fig. 3. Results for different ratios of fin height, h_f , to height of channel, h_c : (a) Nusselt number, Nu, versus Reynolds number, Re; (b) Pressure drop, Δp , across the micro pin fins versus maximum axial velocity, $u_{\max}$.

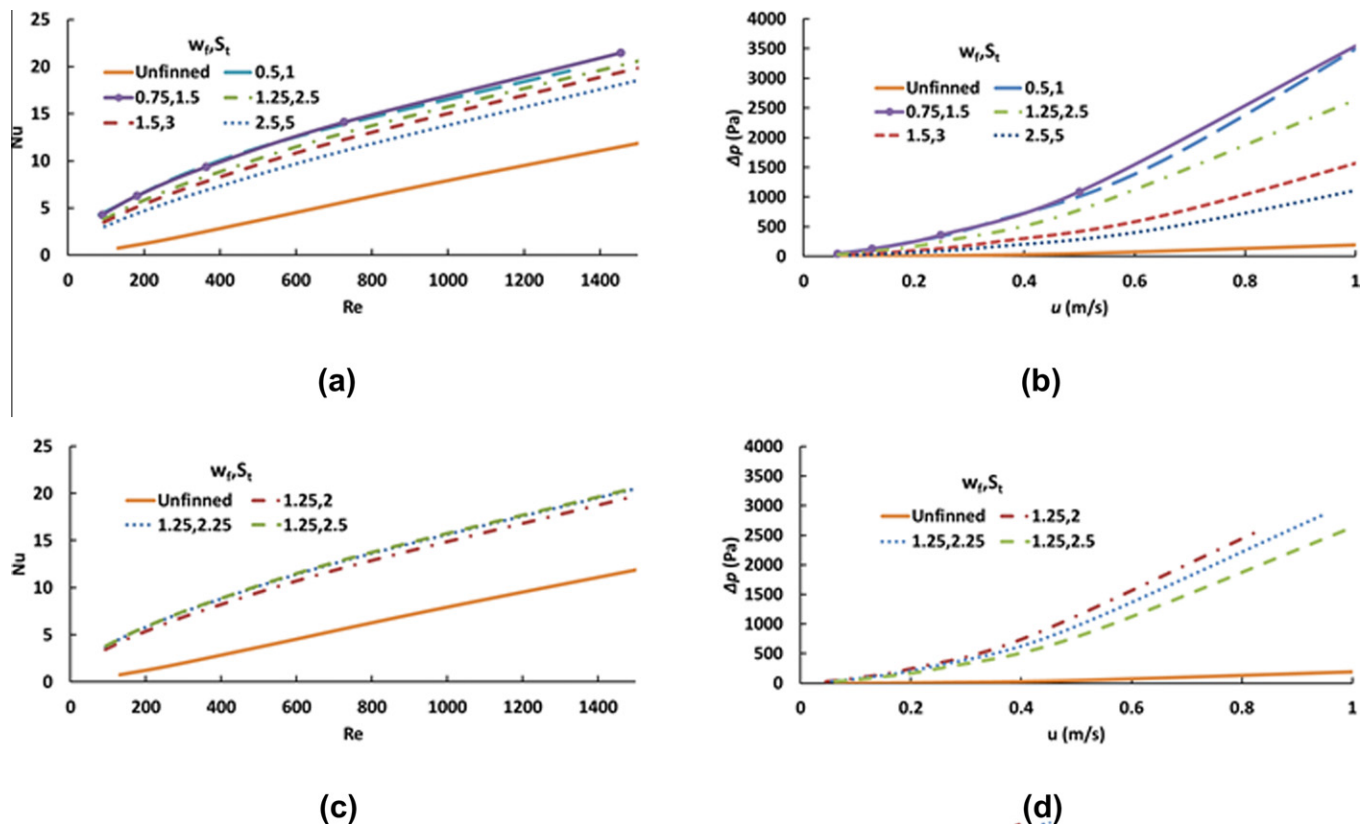


Fig. 4. Results for varying fin width, w_f , and spacing, S_t , and number of fins, N : (a) Nusselt number, Nu , versus Reynolds number, Re , for different fin width and spacing when $2w_f = S_t$; (b) Pressure drop, Δp , versus maximum axial velocity, u_{max} , for different fin spacing and width when $2w_f = S_t$; (c) Nusselt number, Nu , versus Reynolds number, Re , for different fin spacing while width is constant, $w_f = 1.25$ mm; and (d) Pressure drop, Δp , versus maximum axial velocity, u_{max} , for different fin spacing while width is constant, $w_f = 1.25$ mm.

fin with a max pressure drop of 1.4 kPa. The shapes that are more aerodynamic reveal a lower pressure drop because there is less separation of the fluid from the solid body. The opening between the fins disrupts the momentum and the trailing edge of the thermal boundary layer of each oblique fin. This causes the leading edge to re-develop allowing the flow to remain in the developing state. The additions of fins show a big impact in the performance of the heat exchanger.

3.2. Different micro fin height

Fig. 3a and b display the results for different channel clearances. The fin heights are varied from 25% to 75% the length of the channel height. With increasing fin height, the thermal performance increases with increasing Re . From the lowest fin height at 0.25 mm, compared to the unfinned channel, the Nu increase is very high at low Re and the gap decreases to $\approx 58\%$ for high Re . From the largest channel clearance, $h_f = 0.75$ mm, to the smallest, $h_f = 0.25$ mm, there is a Nu increase of $\approx 44\%$. As the fins height is increased, the pressure drop drastically increases. Pressure drop from the largest channel clearance is 0.98 kPa and the smallest channel clearance increases to 7.4 kPa. Because there is a large pressure drop difference for the increasing height and the small Nu relative increase, the suggested fin height is dependent on the available pump parameters used within the cooling apparatus.

3.3. Different micro fin width and spacing

For the third investigation, the fins spacing, width, and number of fins was varied in the channel. Like stated before these parameters are all dependent of each other which made it difficult to

isolate. Therefore to perform this study, a ratio of the total fin width in the x direction of one row to the channel width was held constant. The total length of the square fins occupied in a given row was equal to the total length the fluid flowed through the channel. Initially the spacing between the fins and each row was kept to two times the fin width, $S_t = S_l = 2w_f$, similar to the previous studies. It was discovered that with smaller fin width and spacing, i.e. number of fins was higher, the Nu value was greater than other fins. Fins with width of 0.5 mm projected an 88% higher Nu values than the unfinned channel and 38% higher Nu values than the lowest curve of fins width and spacing at 2.5 mm. Fins with width 2.5 mm still yield a higher Nu value about 50% greater than the unfinned channel with higher Re . This was shown in Fig. 4a. While the smaller fin width and spacing provides higher performance, pressure drop between across the fins is sacrificed. As the

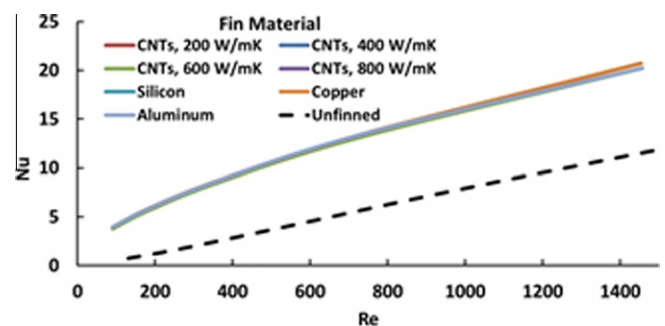


Fig. 5. Nusselt number, Nu , versus Reynolds number, Re , for selected fin material properties.

fin dimensions are reduced, the pressure drop increases. Fins with spacing and width of 1 mm and 0.5 mm the pressure drop is about 15% compared to the bigger fin width and spacing of 2.5 mm and 5 mm and about 4% compared to the unfinned channel. The corresponding graph displaying these results is Fig. 4b. The bigger fins express a smaller pressure drop across the channel. This is probably due to the fewer disturbances the fluid has within the channel. It is interesting to note that for both the pressure drop and the Nu the fins with width of 0.5 mm and 0.75 mm are very similar. The optimum fin width is capped out between these two widths. Similar to the other conclusion, specification of the pump parameters should be taken into account on the number of fins the system can have before the pressure drop is too high.

Once it was determined that smaller fins width and spacing would produce higher thermal performance within the channel, the fin width and number of fins across one row was kept constant at 1.25 mm and 6, respectively, but the spacing between fins varied. As the fin spacing was decreased, to get the most efficient use out of the fins, the columns were centered in the middle of the channel. From the results shown in Fig. 4c and d, as the spacing was closer to equal the fin width, the performance was greater. From the smallest spacing to the largest, i.e. 2–2.5 mm, there was a 6% enhancement for larger fins spacing. With a decrease in fin spacing, the pressure drop increases. There is about 5% increase

in pressure drop from the 2.5 mm spacing to the 2 mm. From the results, it can be concluded that with decrease in width/spacing creates higher thermal performance with a sacrifice in pressure drop. By only changing the spacing, the maximum spacing yields a higher thermal performance providing a minimal pressure drop.

3.4. Different micro fin material properties

In the final study, CNTs, silicon, copper, and aluminum material properties are used as the fin material. The CNTs effective thermal conductivity is varied from 200 to 800 W/mK because that value is still unknown. For single phase flows, changing material properties show little variation within the Nu values, yet, they are still larger than unfinned channel. The results portraying this study are shown in Fig. 5. These findings show that the chosen fin material had little effect on the Nu values. In this study, CNTs are modeled as a solid fin but in reality the fins are more of a porous media that contain small gaps where the fluid can penetrate. The nanotubes can also initiate nucleation sites initiating boiling and therefore enhancing heat transfer. Because of this, the results for the CNT fins are underestimated using this program. More accurate studies need to be conducted in order to verify this assumption made. From this conclusion, the CNTs should yield a greater thermal performance than the other fins. Pressure drops display no difference between the fin

Table 3
MAE values for Nu correlations.

Ref.	Correlation	Scale	MAE
<i>Circle</i>			
Present circle correlation	$Nu_f = 0.08 \left(\frac{S_f}{D_f} \right)^{0.2} \left(\frac{S_f}{D_f} \right)^{0.2} \left(\frac{h_f}{D_f} \right)^{0.25} \left(1 + \frac{dh}{D_f} \right) Re_f^{0.6} Pr^{0.36} \left(\frac{Pr}{Pr_s} \right)^{0.25}$	Mini	9.57
Hwang and Yao [40]	$Nu_f = 0.83 \lambda^{1/3} Re_f^{1/3} Pr^{1/3} \left(\mu_f / \mu_s \right)^{0.14}$ $\lambda = \beta^4 - 1 / (1 + \beta^4) \ln(\beta^2) + (1 - \beta^4)$ $\beta = (1/1 - \xi)^{1/2}; \xi = S_L S_L - A_f / S_L S_L$ $Nu_f = C_1 Re_f^{1/2} Pr^{1/3}$	Macro	37.59
Khan et al. [41]	$C_1 = \left(0.61 \left(\frac{S_f}{D_f} \right)^{0.091} \left(\frac{S_f}{D_f} \right)^{0.053} \right) / \left(\left(\frac{S_f}{D_f} - 1 \right)^{0.5} \left(1 - 2 \exp \left(-1.09 \frac{S_f}{D_f} \right) \right) \right)$	Mini	82.62
Kosar and Peles [42]	$Nu_f = 0.0423 Re_f^{0.99} Pr^{0.21} \left(\frac{Pr}{Pr_s} \right)^{0.25}$	Micro	57.05
Metzger et al. [43]	$Nu_f = 0.135 \left(\frac{S_L - D_f}{D_f} \right)^{-0.34} Re_f^{0.69}$	Macro	7.62
Moore and Joshi [44]	$Nu_f = 0.64 \left(\frac{h_f}{D_f} \right)^{0.36} \left(\frac{dh + h_f}{h_f} \right)^{-0.57} Re_f^{0.64} Pr^{0.36}$	Mini	72.56
Prasher et al. [23]	$Nu_f = 0.281 \left(\frac{S_L - D_f}{D_f} \right)^{-0.63} Re_f^{0.73}$	Micro	61.3
Short et al. [45]	$Nu_f = 0.76 \left(\frac{S_f}{D_f} \right)^{0.16} \left(\frac{S_f}{D_f} \right)^{0.2} \left(\frac{h_f}{D_f} \right)^{-0.11} Re_f^{0.33} Pr^{1/3}$	Macro	45.36
Whitaker [46]	$Nu_f = 2 Re_f^{1/3} Pr^{1/3} \left(\frac{\mu}{\mu_s} \right)^{0.14}$	Macro	71.8
Zukauskas [39]	$Nu_f = 0.9 Re_f^{0.4} Pr^{0.36} \left(\frac{Pr}{Pr_s} \right)^{0.25}$	Macro	59.64
<i>Square</i>			
Present square correlation	$Nu_f = 0.0937 \left(\frac{S_f}{D_f} \right)^{0.2} \left(\frac{S_f}{D_f} \right)^{0.2} \left(\frac{h_f}{D_f} \right)^{0.25} \left(1 + \frac{dh}{D_f} \right)^{0.4} Re_f^{0.6} Pr^{0.36} \left(\frac{Pr}{Pr_s} \right)^{0.25}$	Mini	9.34
Liu et al. [47]	$Nu_f = 0.1245 Re_f^{0.6106} Pr^{0.36} \left(\frac{Pr}{Pr_s} \right)$	Micro	2.1
Liu et al. [47]	$Nu_f = 0.143 Re_f^{0.615} Pr^{1/3}$	Micro	8.9
Qu and Siu-Ho [25]	$Nu_f = 0.0241 Re_f^{0.953} Pr^{0.36} \left(\frac{Pr}{Pr_s} \right)^{0.25}$	Micro	28.76
Qu and Siu-Ho [25]	$Nu_f = 0.0285 Re_f^{0.932} Pr^{1/3}$	Micro	29.67
<i>Diamond</i>			
Present diamond correlation	$Nu_f = 0.036 \left(\frac{S_f}{D_f} \right)^{0.2} \left(\frac{S_f}{D_f} \right)^{0.2} \left(\frac{h_f}{D_f} \right)^{0.25} \left(1 + \frac{dh}{D_f} \right)^{0.4} Re_f^{0.6} Pr^{0.36} \left(\frac{Pr}{Pr_s} \right)^{0.25}$	Mini	8.34
<i>Triangle</i>			
Present triangle correlation	$Nu_f = 0.0454 \left(\frac{S_f}{D_f} \right)^{0.2} \left(\frac{S_f}{D_f} \right)^{0.2} \left(\frac{h_f}{D_f} \right)^{0.25} \left(1 + \frac{dh}{D_f} \right)^{0.4} Re_f^{0.6} Pr^{0.36} \left(\frac{Pr}{Pr_s} \right)^{0.25}$	Mini	9.07
<i>Ellipse</i>			
Present ellipse correlation	$Nu_f = 0.0936 \left(\frac{S_f}{D_f} \right)^{0.2} \left(\frac{S_f}{D_f} \right)^{0.2} \left(\frac{h_f}{D_f} \right)^{0.25} \left(1 + \frac{dh}{D_f} \right)^{0.4} Re_f^{0.6} Pr^{0.36} \left(\frac{Pr}{Pr_s} \right)^{0.25}$	Mini	9.32
<i>Hexagon</i>			
Present hexagon correlation	$Nu_f = 0.0752 \left(\frac{S_f}{D_f} \right)^{0.2} \left(\frac{S_f}{D_f} \right)^{0.2} \left(\frac{h_f}{D_f} \right)^{0.25} \left(1 + \frac{dh}{D_f} \right)^{0.4} Re_f^{0.6} Pr^{0.36} \left(\frac{Pr}{Pr_s} \right)^{0.25}$	Mini	9.47

materials because the geometry remained the same for this section of the study.

Pressure drop for these studies have shown critical in optimizing fin geometry and topology. For a single phase, laminar, flow through a minichannel, Nu values for pin fins did show improvement compared to the unfinned channel; however, various shapes, heights, widths, spacing and thermal properties only show slight deviation.

4. Mathematical correlations

To further understand our numerical data, empirical relationships between (1) Nu_f and Re_f and (2) u_{max} and Δp were established. This was completed in order to be able to predict the correlations based on their geometrical parameters. Expressions for Nu_f and f were developed for this study. Once they were established, they were compared to other pin fin studies that have developed expressions for Nu_f and f . These expressions were obtained based on the first two studies: varying geometries and heights. Our correlations were then applied to a specific example and results are presented in terms of Nu_f and Δp .

4.1. Heat transfer

The conventional correlation for Nu_f for flow through a tube bank is typically in the form:

$$Nu_f = C_{Nu} Re_f^X Pr^{1/3} \quad (8)$$

developed by Zukauskas [39], where C_{Nu} and X are constants and Pr is the Prandtl number corresponding to the fluid. Because the height and shape of the fins and the clearance of above the fins are a major focus in this paper, the new expression incorporates more terms to further define the geometry of the fins: S_L , S_t , h_f , and the channel

clearance, dh . From the data collected, using a best fit curve the unknown parameters were obtained for new Nu_f correlations for each shape. The new expressions:

$$Nu_f = C_{Nu} \left(\frac{S_L}{D_f}\right)^S \left(\frac{S_t}{D_f}\right)^U \left(\frac{h_f}{D_f}\right)^V \left(1 + \frac{dh}{D_f}\right)^W Re_f^X Pr^Y \left(\frac{Pr}{Pr_s}\right)^Z \quad (9)$$

C_{Nu} , S , U , V , W , X , Y , and Z are all constants determined from the numerical data. These correlations were compared to others developed by previous studies [23,25,39–47]. For each shape, the expression differs only with the constant, C_{Nu} . The exponents are all the same. This is shown as:

$$Nu_f = C_{Nu} \left(\frac{S_L}{D_f}\right)^{0.2} \left(\frac{S_t}{D_f}\right)^{0.2} \left(\frac{h_f}{D_f}\right)^{0.25} \left(1 + \frac{dh}{D_f}\right)^{0.4} Re_f^{0.6} Pr^{0.36} \left(\frac{Pr}{Pr_s}\right)^{0.25} \quad (10)$$

for

Geometry	Circle	Square	Diamond	Triangle	Ellipse	Hexagon
C_{Nu}	0.08	0.0937	0.036	0.0454	0.0936	0.0752

Re_f for these correlations are determined based on the hydraulic diameters of the fins, D_f , and the average velocity flowing through the fins. Re_f and D_f are defined in Eqs. (11) and (12), respectively.

$$Re_f = \frac{\rho u D_f}{\mu} \quad (11)$$

$$D_f = \frac{4A_f}{P_f} \quad (12)$$

A_f is the cross-sectional area of the fin and P_f is the perimeter. These expressions are effective for Re_f in the laminar regime.

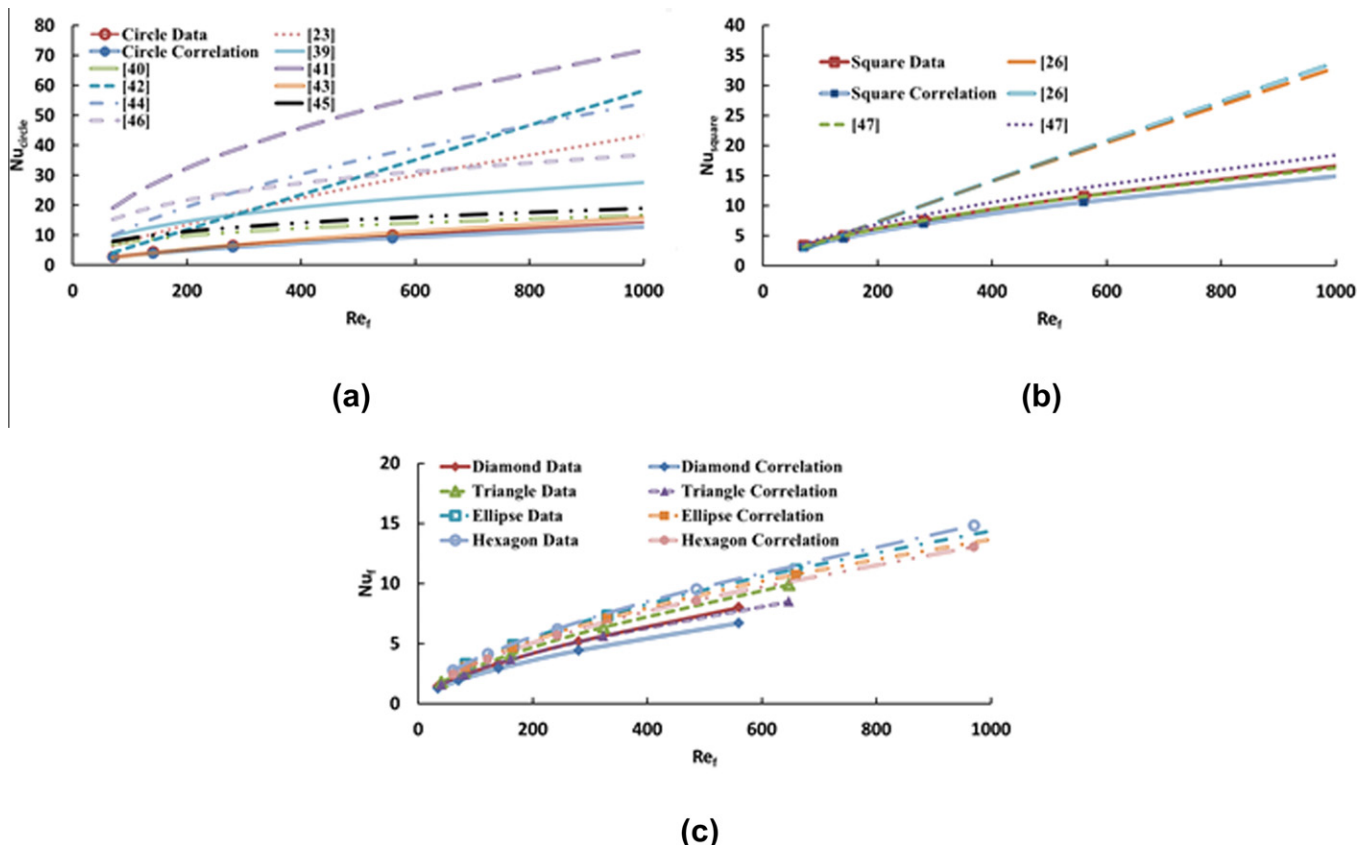


Fig. 6. Nusselt number, Nu_f , versus Reynolds number, Re_f , for numerical data compared to newly developed and other correlations: (a) Using circular shaped pin fin; (b) Using square shaped pin fin; and (c) Using diamond, triangle, ellipse, and hexagonal shaped pin fins.

These results are compared to other generated correlations using the mean absolute error, MAE, defined in Eq. (13) [24,25,28,48]. This equation is an average error between the data received from the computational runs and the data predicted from the correlation provided. The values with the smallest MAE express the better correlation.

$$\text{MAE} = \frac{1}{M} \sum_{i=1}^M \frac{|data - data_{pred}|}{data_{pred}} \times 100\% \quad (13)$$

The parameter M is the number of velocities used in the study. Table 3 shows the Nu_f correlations for the corresponding references and the MAE value with respect to the numerical data obtained. It is separated into the geometrical shapes in which the correlations were developed from. The comparisons can also be further seen in Fig. 6. From Table 3, the MAE for the correlations developed for each shape is under 10. Liu et al. [47], Metzger et al. [43], and Qu and Siu-Ho [25] show decent error values closest

to the numerical data with values of 2.1/8.9, 7.62, and 28.76/29.67, respectively. Correlations developed using micro pin fins versus the conventional macro scales fins in general had smaller MAE values. With the exception of the Metzger et al. [43] correlation, conventional correlations are not adequate to predict the heat transfer across a micro pin fin array in a minichannel. Overall the new Nu_f correlation obtained is a good representation for our data.

4.2. Pressure drop

Pressure drop across a tube bank can be obtained using the expression:

$$\Delta p = N_y f \frac{\rho u_{\max}^2}{2} \quad (14)$$

After reviewing many relations from other studies to obtain f , the correlation developed for each shape are in the form defined by Eq. (15), used by Short et al. [49]. This expression is defined in

Table 4
MAE values for f correlations.

Ref.	Correlations	Scale	MAE
<i>Circle</i>			
Present circle correlation	$f = 2.963 \left(\frac{S_f}{D_f}\right)^{0.2} \left(\frac{h_f}{D_f}\right)^{0.18} \left(1 + \frac{dh}{D_f}\right)^{0.2} Re_f^{-0.435}$	Mini	7.06
Armstrong and Winstanley [50]	$f = 2.06 \left(\frac{S_f}{D_f}\right)^{-1.1} Re_f^{-0.16}$	Macro	29.35
Chilton and Generaux [51]	$f = \frac{106}{Re_f}$	Macro	54.47
Gaddis and Gnielski [52]	$f = 280\pi \left(\left(\left(\frac{S_f}{D_f}\right)^{0.5} - 0.6 \right)^2 + 0.75 \right) / Re_f \left(4 \frac{S_f}{D_f} - \pi \right) C^{16}$ $C = \frac{S_f}{D_f}$ for $\frac{S_f}{D_f} \geq \frac{1}{2} \sqrt{2 \frac{S_f}{D_f} + 1}$ $C = \frac{S_f}{D_f}$ for $\frac{S_f}{D_f} < \frac{1}{2} \sqrt{2 \frac{S_f}{D_f} + 1}$	Macro	279.5
Gunther and Shaw [53]	$f = \frac{180}{Re_f} \left(\frac{4S_f}{\pi D_f^2} - 1 \right)^{0.4} \left(\frac{S_f}{D_f} \right)^{0.6}$	Macro	64.22
Kast [54]	$f = \left(\frac{128}{Re_f} + \frac{4}{Re_f^{0.16}} \right)$	Macro	85.87
Kosar et al. [28]	$f = \pi_1 + \pi_2$ $\pi_1 = \frac{1739}{Re_f^{1.07}} \left(\frac{h_f/D_f}{h_f/D_f + 1} \right)^{11} \left(\frac{S_f}{A_f} \right)^{-0.3}$ $\pi_2 = \frac{345}{Re_f^1} \left(\frac{1}{h_f/D_f + 1} \right)^{2.0} \left(\frac{S_f}{A_f} \right)^{-0.3}$	Micro	60.33
Moores and Joshi [44]	$f = 19.04 \left(\frac{h_f}{D_f} \right)^{-0.742} \left(\frac{dh+hc}{hc} \right)^{0.505} Re_f^{-0.502}$	Mini	87.87
Moores et al. [55]	$f = 10.52k_1 \left(\frac{h_f}{D_f} \right)^{0.28+(1-k_1)} Re_f^{-0.39+(1-k_2)} k_1 = \exp^{4.3} (dh/h_f)$ $k_2 = \exp^{0.8} (dh/h_f)$	Mini	79.98
Prasher et al. [23]	$f = 3.5 \left(\frac{h_f}{D_f} \right)^{0.724} \left(\frac{S_f-D_f}{D_f} \right)^{-0.442} \left(\frac{S_f-D_f}{D_f} \right)^{-0.245} Re_f^{-0.58}$	Micro	16.93
Short et al. [49]	$f = 140.4 \left(\frac{S_f}{D_f} \right)^{-1.3} \left(\frac{S_f}{D_f} \right)^{-0.78} \left(\frac{h_f}{D_f} \right)^{-0.55} Re_f^{-0.65}$	Macro	72.43
<i>Square</i>			
Present square correlation	$f = 5.28 \left(\frac{S_f}{D_f}\right)^{0.2} \left(\frac{S_f}{D_f}\right)^{0.2} \left(\frac{h_f}{D_f}\right)^{0.18} \left(1 + \frac{dh}{D_f}\right)^{0.2} Re_f^{-0.435}$	Mini	6.82
Konishi et al. [56]	$f = 10.485 Re_f^{-0.35}$	Micro	63.94
Konishi et al. [57]	$f = \frac{55,631}{Re_f} + \frac{2,1114}{Re_f^{0.09597}}$	Micro	62.61
Liu et al. [47]	$f = 1.6361 Re_c^{-0.01076} E^{-0.94496} \quad E = \frac{2S_f}{W_c / \sin(45^\circ)}$	Micro	39.12
Qu and Siu-Ho [26]	$f = 20.09 Re_f^{-0.547}$	Micro	37.7
<i>Diamond</i>			
Present diamond correlation	$f = 1.81 \left(\frac{S_f}{D_f}\right)^{0.2} \left(\frac{S_f}{D_f}\right)^{0.2} \left(\frac{h_f}{D_f}\right)^{0.18} \left(1 + \frac{dh}{D_f}\right)^{0.2} Re_f^{-0.435}$	Mini	6.67
Kosar et al. [28]	$f = \pi_1 + \pi_2$ $\pi_1 = \frac{1126}{Re_f^1} \left(\frac{h_f/D_f}{h_f/D_f + 1} \right)^{1.5} \left(\frac{S_f}{A_f} \right)^{-0.4}$ $\pi_2 = \frac{6.6}{Re_f^1} \left(\frac{1}{h_f/D_f + 1} \right)^{1.7} \left(\frac{S_f}{A_f} \right)^{-1}$	Micro	64.78
<i>Triangle</i>			
Present triangle correlation	$f = 2.45 \left(\frac{S_f}{D_f}\right)^{0.2} \left(\frac{S_f}{D_f}\right)^{0.2} \left(\frac{h_f}{D_f}\right)^{0.18} \left(1 + \frac{dh}{D_f}\right)^{0.2} Re_f^{-0.435}$	Mini	5.98
<i>Ellipse</i>			
Present ellipse correlation	$f = 3.44 \left(\frac{S_f}{D_f}\right)^{0.2} \left(\frac{S_f}{D_f}\right)^{0.2} \left(\frac{h_f}{D_f}\right)^{0.18} \left(1 + \frac{dh}{D_f}\right)^{0.2} Re_f^{-0.435}$	Mini	6.62
<i>Hexagon</i>			
Present hexagon correlation	$f = 4.43 \left(\frac{S_f}{D_f}\right)^{0.2} \left(\frac{S_f}{D_f}\right)^{0.2} \left(\frac{h_f}{D_f}\right)^{0.18} \left(1 + \frac{dh}{D_f}\right)^{0.2} Re_f^{-0.435}$	Mini	9.39

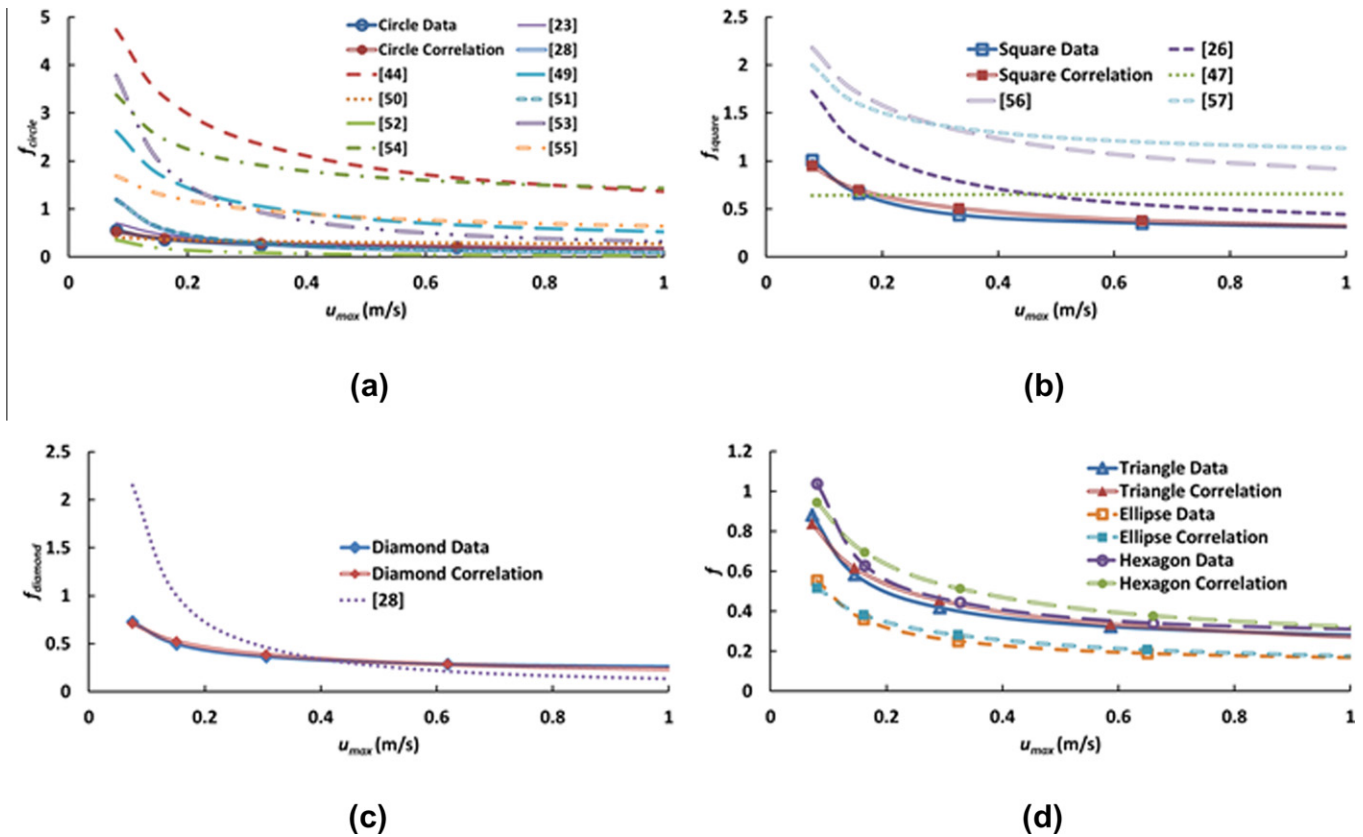


Fig. 7. Pressure drop, Δp , versus maximum axial velocity, u_{max} , for numerical data compared to newly developed and other correlations: (a) Using circular shaped pin fin; (b) Using square shaped pin fin; (c) Using diamond shaped pin fins; and (d) Using triangle, ellipse, and hexagonal shaped pin fins.

terms of many geometrical parameters outlining the staggered formation similar the Nu_f equation in the previous section; therefore, could reflect this data best.

$$f = C_f \left(\frac{S_L}{D_f} \right)^S \left(\frac{S_t}{D_f} \right)^U \left(\frac{h_f}{D_f} \right)^V \left(1 + \frac{dh}{D_f} \right)^W Re_f^X \quad (15)$$

C_f , S , U , V , W and X are all constants determined from the data. This expression incorporates the varying height, the location (S_L and S_t) of the fins within the staggered array and the clearance above the fins for the fluid to flow. The exponents for each shape are constant with only a different C_f and are shown in the next equation.

$$f = C_f \left(\frac{S_L}{D_f} \right)^{0.2} \left(\frac{S_t}{D_f} \right)^{0.2} \left(\frac{h_f}{D_f} \right)^{0.18} \left(1 + \frac{dh}{D_f} \right)^{0.2} Re_f^{-0.435} \quad (16)$$

for

Geometry	Circle	Square	Diamond	Triangle	Ellipse	Hexagon
C_f	2.96	5.28	1.81	2.45	3.44	4.53

Re_f is determined based on the hydraulic diameters of the fins and the maximum velocity, u_{max} , flowing through the fins. This expression is effective for Re_f in the laminar regime.

Table 4 compares the MAE values for the correlations of f developed for each shape with other existing correlations. Each of the correlations developed have a MAE less than 10 for all shapes which displayed better values than the others. The expression developed by Prasher et al. [23] showed the next closest in

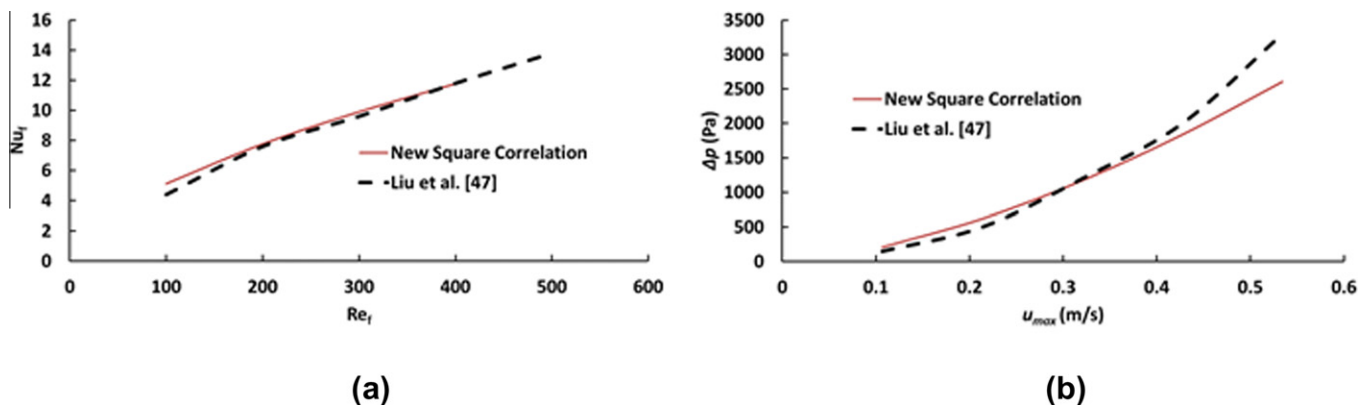


Fig. 8. Numerical validation of the new correlations with the Liu et al. [47] experimental data: (a) Nusselt number, Nu_f , versus Reynolds number, Re_f ; (b) Pressure drop, Δp , versus maximum axial velocity, u_{max} .

determining the numerical data with an MAE value of 16.93 and the next Liu et al. [47] and Qu and Siu-Ho [26] with about 40. This is shown in Fig. 7. It seems as the MAE values closer resemble expressions developed for microchannels versus conventional scaled channels with fins.

4.3. Validation of correlations

To validate the new correlations, experimental results taken from work completed by Liu et al. [47] for a heat sink containing 625 square micro pin fins of $445 \times 445 \mu\text{m}^2$ in a staggered array was solved with water used as the working fluid flowing with Re ranging from 60 to 800. The fins and channel have a height of 3 mm. The longitudinal and transverse spacing of the fin measured $565.7 \mu\text{m}$. The ambient fluid temperature is initially at room temperature and heat is applied to the bottom surface. The geometry of the fins fit the limits of the new equations developed and the Nusselt number, Nu_f , and pressure drop, Δp , across the finned structure were calculated using new correlations and compared to the experimental data. This is shown in Fig. 8. Based on the example, both correlations developed for Nu_f and f are similar to that of the value obtained in the experimental work within 15%. This proves that the new expression for the Nu_f can accurately predict the heat transfer through the channel with micro pin fins on the surface. These correlations obtained can provide suitable values for the pressure drop and the Nu_f across a staggered finned array.

5. Conclusion

An optimization study was completed in order to understand the effects of micro fins on the surface of a minichannel to obtain maximum thermal performance. From previous studies, use of microstructures on the surfaces of micro/minichannels has been effective with increased heat removal from a heated surface. Four parametric studies have been conducted in order to optimize the fins: (1) using different geometrical shaped fins – circle, square, triangle, ellipse, diamond, and hexagon placed on the bottom surface of a minichannel in a staggered formation, (2) varying fin to channel height ratio in a square pin fin geometry formation, (3) varying fin width and spacing, and (4) defining square pin fins as CNTs, silicon, copper, and aluminum. In addition to those studies, Nusselt number and Darcy friction factor correlations were then obtained to fit our data and compared to previous studies. This work was completed by varying inputted velocities and constant heat fluxes while keeping the fluid in the single phase, laminar flow regime. The following conclusions can be drawn from the results:

- Fin shapes yield Nu values within 37% of each other at high Re with the triangle fins as the highest and the circle and ellipse fins lower; however, the pressure drop values are greatly affected. The ellipse and circle fin portray the smallest pressure drop value because of its aerodynamic shape with the lowest Nu values larger than 45% when compared to the unfinned channel.
- With increasing fin height the Nu increases; however, the pressure drop also increases drastically. From the lowest tested fin height to channel ratio to the highest, the difference in Nu is 44% and the pressure drop differential at higher Re is almost 7.4 kPa. Fin height to channel height ratio should be determined based on the maximum pressure drop allowed for the pump used.
- With decreasing fin width and spacing the Nu increases; however, the increase in pressure drop is significant. For fin width at 0.5 mm there was an increase in performance is 88% higher than that of the unfinned channel and about 38% higher Nu values to the larger fin width and spacing. Pressure drop increased significantly with a 15% increase to the unfinned channel.
- In keeping the fin width and the number of fins constant, the fins spacing was varied. The results showed that allowing for the maximum spacing, or the spacing equal to the fin width, the thermal performance is better by 6% with minimal pressure drop of about 5%.
- Fin material shows little effect on fin performance; however, fins with CNTs are not modeled as a porous media which is expected to increase the surface area that the fluid interacts with and can hasten nucleate boiling onset which could increase heat transfer performance significantly. Therefore the thermal performance for the CNT fins is under predicted.
- Empirical expressions for both Nu_f and f were provided and validated for all six shapes and compared to other existing correlations. The expressions were curve fit to the data obtained and when creating these expressions, the only value that varied between the different shapes was the constant.

More intensive studies need to be conducted in accurately modeling CNTs as the porous media. Triangular fins with larger fin height, smaller fin width, and spacing double the fin width maximizes the number of fins in each row and yields better performance. The digital revolution is upon us, but without significant gains in nano/micro/mini heat transfer technology, the new advancements will be unattainable.

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